IIBS Computer Lab Management System

Offline LAN Deployment, Exam Internet Cutoff & Administration Manual

1. System Overview

This software suite is tailored for institutional computer labs operating over pure LAN (no external internet required for operations). It provides student self-registration, credential-based desktop unlock, real-time activity monitoring, 1-hour idle auto-shutdown, one-shot broadcast file sharing, administrative password resets, and **1-Click Exam Mode** to block external internet across all machines while keeping desktop functionality.

Component	Function & Mechanism
Master Server & Portal	Runs on Teacher PC (Flask + SQLite). Exposes web dashboard on port 5000 for monitoring, file sharing, password resets, and exam toggles.
Client Kiosk Agent	Runs on student PCs. Fullscreen lock screen on boot. Unlocks upon student authentication into a lightweight session widget.
Exam Internet Cutoff	Blocks outbound HTTP/HTTPS/DNS (80, 443, 53) via Windows Firewall while whitelisting the Master Server IP. Allows normal desktop apps for exams.
Idle Auto-Shutdown	Tracks native Windows mouse/keyboard activity. If idle exceeds 60 minutes, system safely initiates shutdown to save energy.
Audit & System Logs	Records all logins, failed attempts, exam toggles, and shutdowns. Full CSV export available in 1 click.

2. Step 1: Teacher Master Server Setup

1. Obtain the Teacher PC's LAN IP address by running ipconfig in Command Prompt (e.g. 192.168.1.100).
2. Navigate to d:\Antigravity\IIBS\lab_manager\ and double-click **start_server.bat**.
3. Open any web browser on your PC or mobile device on the same LAN and navigate to:

```
http://127.0.0.1:5000/admin (or http://<TEACHER_IP>:5000/admin)
```

3. Step 2: Student Lab PC Setup

1. Open lab_manager\client\client_config.json and set your Teacher's LAN IP address:

```
{ "server_url": "http://192.168.1.100:5000" }
```

2. Launch **start_client.bat** on the student machine.
3. The client locks into full-screen mode. New students click **New Student Register**, fill Roll Number, Full Name, Username, and Password, then switch to the **Student Login** tab.
4. Upon login, the screen unlocks into normal desktop view with a small floating session widget.

4. Step 3: Conducting Exams (Disabling Internet in Normal Mode)

During lab examinations, instructors often need students to have normal access to local applications (IDE, Word, compilers, or local intranet exam portals) while ensuring external internet (Google, ChatGPT, web browsing) is strictly blocked.

• **To Start an Exam:** Click **'Enable Exam Mode (Block Student Internet)'** on the Admin Web Portal. All student PCs automatically apply Windows Firewall rules blocking web ports (80, 443) and DNS (53) while keeping LAN access to the teacher PC intact. **The LAB MASTER PC (Teacher Server) is strictly EXCLUDED and maintains its full internet access.** The student PC displays a red [EXAM IN PROGRESS - NO INTERNET] badge.

• **To End the Exam:** Click **'End Exam & Restore Internet'**. The firewall rules are removed instantly and full student internet is restored.

5. Step 4: System Logs & Student Password Reset

• **Password Resets:** Under 'Student Records', search any student by roll number or username and click **Reset Password**. The student can immediately log in on their lab PC with the new password.

• **Audit & System Logs:** Every registration, successful/failed login, exam cutoff event, and shutdown is timestamped and logged. Click **'Export Logs (CSV)'** to download an Excel-ready audit file.

6. Step 5: Removing Student Admin Rights & Windows Lockdown

To prevent students from bypassing the kiosk, killing tasks, or tampering with network settings, configure Windows as follows:

1. **Standard User Account:** Go to *Windows Settings* > *Accounts* > *Other users*. Create a local account named Student and ensure account type is set to **Standard User** (not Administrator).

2. **Disable Task Manager:** Press Win + R, type gpedit.msc. Navigate to *User Configuration* > *Administrative Templates* > *System* > *Ctrl+Alt+Del Options*. Set **Remove Task Manager** to **Enabled**.

3. **Configure Auto-Start:** Press Win + R, type shell:startup. Create a shortcut to start_client.bat (or client.exe) inside this folder so the lock screen loads automatically on boot.

7. Step 6: Compiling Standalone Executable (.exe)

To deploy on student PCs without needing Python installed, compile the client into a single .exe file:

```
pip install pyinstaller cd lab_manager\client pyinstaller --onefile --noconsole client.py
```

Copy the generated dist\client.exe and client_config.json to each student PC.